

# Transcriptome Assemblies

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# Acknowledgement Of Country

I'd like to acknowledge the Kaurna people as the traditional owners and custodians of the land we know today as the Adelaide Plains, where I live & work.

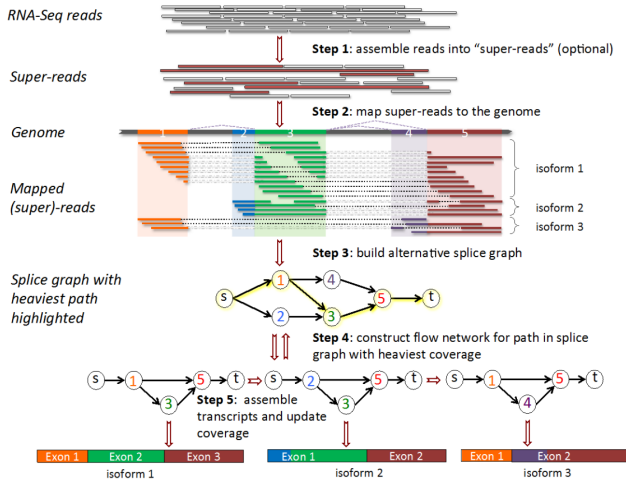
I also acknowledge the deep feelings of attachment and relationship of the Kaurna people to their place.

I pay my respects to the cultural authority of Aboriginal and Torres Strait Islander peoples from other areas of Australia, and pay my respects to Elders past, present and emerging, and acknowledge any Aboriginal Australians who may be with us today

# Transcript Assembly

- Beyond (or as part of) differential expression analysis  $\implies$  *transcriptome assembly*
- Well annotated genomes have gene models defined
  - May only have a reference from related organism
- Biology can be messy & unexpected
  - Unexpected cryptic transcripts
  - Novel TSS, UTRs etc
  - *How well do existing annotations describe our samples?*
- We can assemble *de novo* or using *reference guided* strategies

# Reference Guided Assembly



- With a good reference genome  $\Rightarrow$  *StringTie* can identify novel transcripts
- Un-annotated genes/lncRNA
- Relies on a *splice-graph* to assemble transcripts
- High expression  $\Rightarrow$  High confidence assembly
- Low expression  $\Rightarrow$  Less confidence

Figure 1: Image taken from (Pertea et al. 2015)

# Reference Guided Assembly

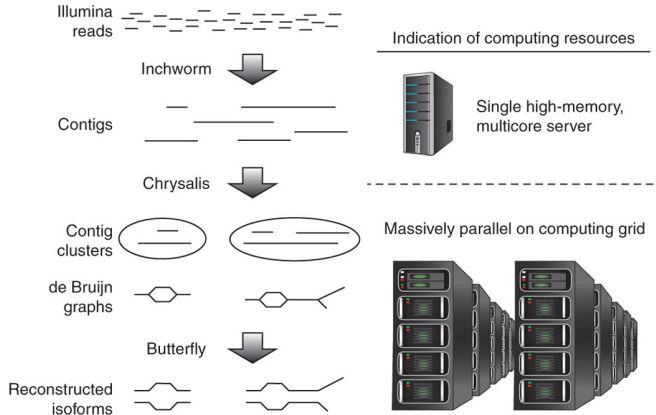
- StringTie returns a GTF with novel transcripts added to reference (See [here](#))
  - Gene & Transcript annotations capture the entire transcribed region
  - Exons are annotated by transcript + gene
- Can merge assemblies across libraries/samples
  - Use merged GTF to obtain counts with `featureCounts` etc
  - Also returns counts for DE analysis

# Full Transcriptome Assembly



- Complete *de novo* transcriptome assembly using Trinity (Grabherr et al. 2011)
  - Far more computationally demanding than *StringTie*
  - Can also perform a *reference-guided assembly*
- Best option where no reference genome is available
  - Or reference is low quality
- Will be tissue specific  $\implies$  a subset of transcripts will be assembled
- Returns a fasta file naive to any reference genome
  - Gene/Transcript clusters in sequence header

# Full Transcriptome Assembly



**Figure 2:** Figure from Haas et al. (2013)

## 1 Inchworm

- Naively assembles reads into contigs

## 2 Chrysalis

- Pools contigs into *de Bruijn* graph

## 3 Butterfly

- Trims *de Bruijn* graph and compares against raw reads

# De Bruijn Graphs

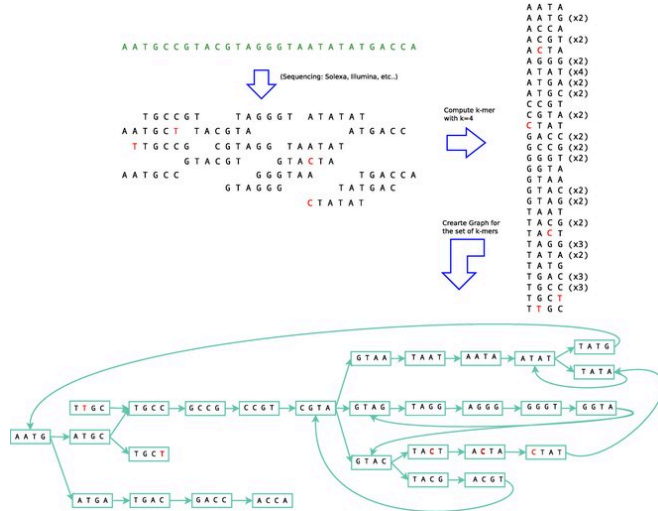


Figure 3: An example de Bruijn graph (Wikimedia commons)

- Reads represent fragmented transcripts
- Reads broken into shorter  $k$ -mers
- Adjacent  $k$ -mers  $\rightarrow$  graph
- Wise choice of  $k$  helps extend graph well beyond read length
- Different  $k \implies$  different assembly
- Well-represented paths through graph  $\rightarrow$  transcripts



# A Trinity Assembled Transcriptome

- Assembly returns a Fasta file with all assembled transcripts

```
>TRINITY_DN1000_c115_g5_i1 len=247 path=[31015:0-148 23018:149-246]  
AATCTTTTTTGGTATTGGCAGTACTGTGCTCTGGGTAGTGATTAGGGCAAAAGAAGACAC  
ACAATAAAGAACCAGGTGTTAGACGTCAGCAAGTCAAGGCCTTGTTCTCAGCAGACAGA  
AGACAGCCCTTCTCAATCCTCATCCCTTCCCTGAACAGACATGTCTTCTGCAAGCTTCTC  
CAAGTCAGTTGTTTACAGGAACATCATCAGAATAAATTTGAAATTATGATTAGTATCTGA  
TAAAGCA
```

- Read cluster TRINITY\_DN1000\_c115; Gene g5; Isoform i1
- Pathway through graph is output
- If diploid organism → both maternal & paternal transcripts

# Assessing Assembly Quality

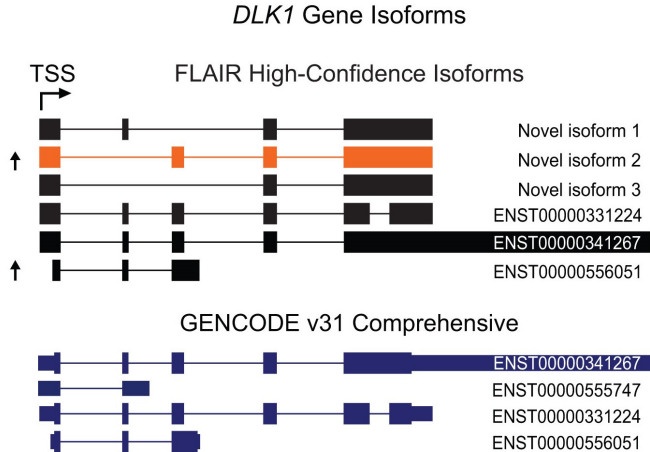
- Benchmarking Universal Single-Copy Orthologs (BUSCO) (Tegenfeldt et al. 2025)
  - Checks expected genes conserved across species
  - Assess assembly quality against expected true positives
  - Most (but not all) orthologs expressed in assembled tissue
- Example BUSCO output: **C:89.0%[S:85.8%,D:3.2%],F:6.9%,M:4.1%,n:3023**
  - **C**: Complete orthologs
    - **S**: Single copy + **D**: Duplicated copy
  - **F**: Fragmented copy
  - **M**: Missing copy
  - **n**: Total orthologs

# Long Read Technology

# Long Read Technology

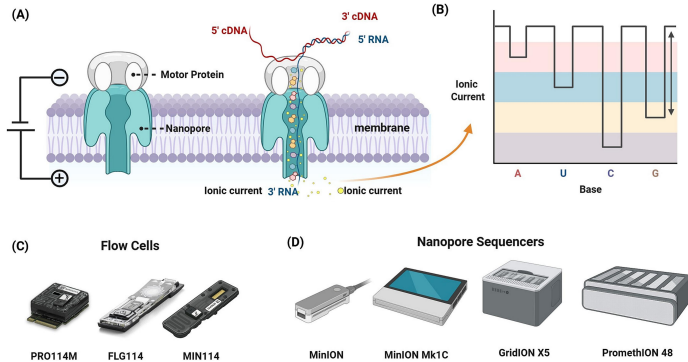
- Most transcriptome assemblies performed using Trinity/StringTie
  - Illumina reads  $\leq 2 \times 150$ nt
- Long Reads are becoming dominant for assemblies
  - Sequence (near-)complete transcripts
- Capture “difficult” transcripts:
  - Transposable Elements (TEs)
  - *DUX4* within repeat array with 11-150 repeats
- Quantification approaching short-read consistency
  - Overcomes fragmentation bias in short-read technology (Chen et al. 2025)

# Long Read Technology



**Figure 4:** Image from Gleeson et al. (2021). Isoforms observed in SH-SY5Y cells

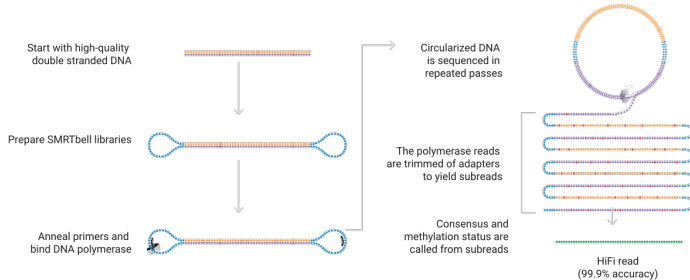
# Oxford Nanopore (ONT)



- From 50 bp to >4 Mb
- Can directly sequence RNA
  - No PCR bias
  - RNA-modifications
- Also cDNA sequencing
- Errors corrected using deep learning

Figure 5: Image from Sun et al. (2025)

# Pacific Biosciences (PacBio)



- PacBio IsoSeq:
  - Up to 25kb
  - Sequence cDNA only
  - Circularisation  $\Rightarrow$  Highly accurate reads

**Figure 6:** Image from

<https://www.pacb.com/technology/hifi-sequencing/how-it-works/>

## References



- Chen, Ying, Nadia M Davidson, Yuk Kei Wan, Fei Yao, Yan Su, Hasindu Gamaarachchi, Andre Sim, et al. 2025. "A Systematic Benchmark of Nanopore Long-Read RNA Sequencing for Transcript-Level Analysis in Human Cell Lines." *Nat. Methods* 22 (4): 801–12.
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